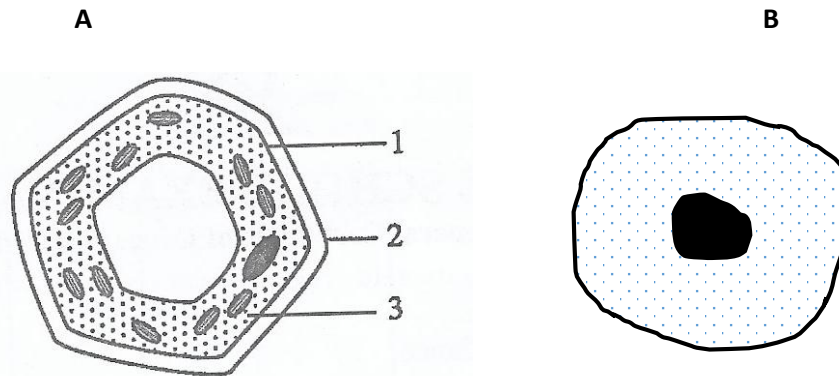


SECTION A

ANSWER **ALL** QUESTIONS

1. a) The diagrams below show two cells **A** and **B**.



- i) Name the cells **A** and **B** above.

Cell A.....

Cell B.....[2]

- ii) State an instrument that can be used to view the cells **A** and **B**.

.....[1]

- iii) Outline any **two** similarities and any **two** differences, shown on the diagrams, between the cells above.

Similarity **1**.....

.....

Similarity **2**.....

.....

Differences

Cell A	Cell B
1.	
2.	

[4]

- b) State any **one function** of **each** of the following parts of the cells above

Part	Function
i) Nucleus	
ii) Cytoplasm	
iii) Cell membrane	

[3]

- c) Outline any four differences between living and non-living organisms.

Living organisms	Non-living organisms
i)	
ii)	
iii)	
iv)	

[4]

2. a) i) Define the term Balanced Diet.

.....

[2]

- ii) List any **three** main sources of protein in the human diet.

1.....
 2.....
 3.....

[3]

- iii) Explain the importance of dietary fibre in the human body.

.....

[2]

- b) i) Complete the table below about the composition of air.

composition	% in air
Oxygen	21
	0.03
Moisture	
	1

[4]

- ii) From the completed table above, state the **two** respiratory gases.

1.....
 2.....[2]

- iii) Briefly, describe how you can test for the presence of **each** of the respiratory gases you have stated in b)ii) above.

Respiratory gas	Test
1.	
2.	

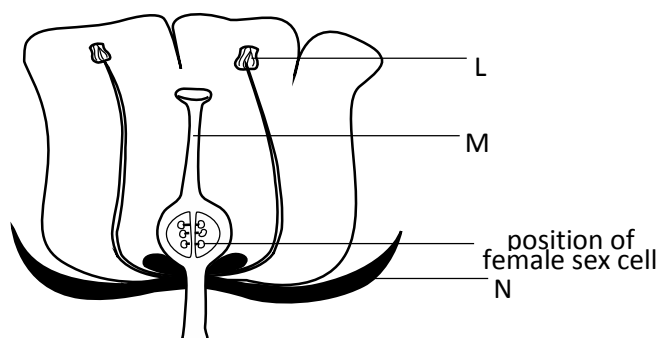
[4]

d) Differentiate between Breathing and Respiration.

Breathing.....
[2]

Respiration.....
[2]

3. a) The diagram below shows a simple flower



i) In the diagram above and label the parts **L**, **M** and **N**.

L.....
M.....
N.....[3]

ii) State the functions of the parts **L** and **N**.

Function of **L**.....

 Function of **N**.....
[2]

ii) State where fertilisation takes place in the flower above.

.....[1]

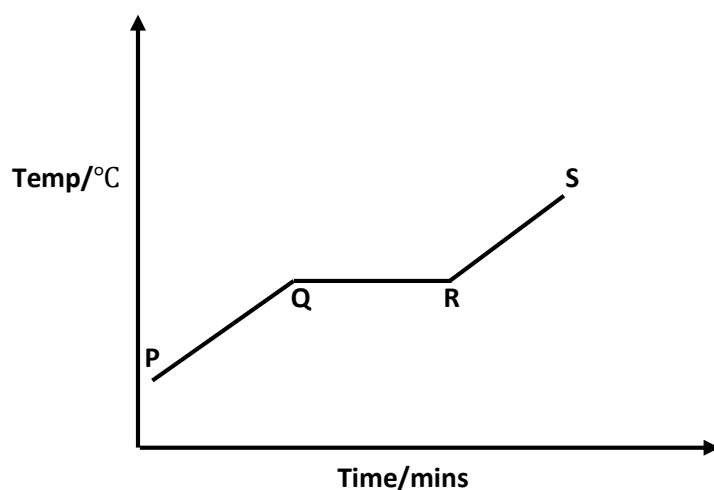
b) i) Define the term puberty.

.....
[2]

iii) Describe any four secondary sexual characteristics in girls.

- 1.....
- 2.....
- 3.....
- 4.....[4]

4. a) The diagram below shows a graph obtained when solid naphthalene was heated and the change in temperature was measured against time.



i) Name the graph above.
.....[1]

ii) In what state of matter is naphthalene in the regions **PQ** and **RS**?

PQ.....
RS.....[2]

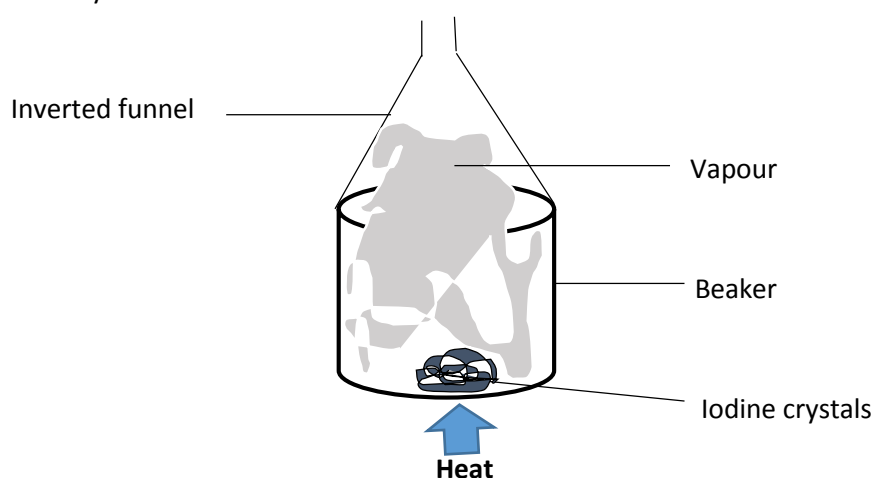
iii). Use the kinetic Theory Matter to describe what is happening between **Q** and **R**.

.....
.....
.....
.....
.....[4]

- b) Draw a diagram of the apparatus you need to carry out an experiment to come up with the graph in a) above.

[5]

- c) The diagram below shows an experiment to demonstrate the change of state when solid iodine crystals are heated.



- i) Give the colours of the iodine crystals and the vapour produced.

Iodine crystals.....

Vapour.....[2]

- ii) State and describe the process shown in the diagram above.

Process	Description

[3]

- iii) Explain why the experiment should be carried out in a fume cupboard.

.....

.....

.....

.....[2]

SECTION B

Answer **all** questions

5. a) What is meant by the term **Health**?

.....

.....

.....

.....[2]

- b) State three methods of disease transmission.

1.....

2.....

3.....[3]

- c) Describe three methods of disposing litter.

1.....

2.....

3.....[3]

6. a) State and explain any three rules of the laboratory.

Rule	Explanation
1.	
2.	
3.	

[6]

- b) Define the following terms

i) Matter.....

.....

ii) Atom.....

.....

iii) Element.....

.....[6]

- c) i) Outline four differences between a compound and a mixture.

Compound	Mixture
1.	
2.	
3.	
4.	

[4]

- ii) Using diagrams, describe how you can separate a mixture of sand and table salt.

[5]

7. a) Outline any **five** properties of acids.

- 1.....
- 2.....
- 3.....
- 4.....
- 5.....

[5]

- b) You are given the following:

Sodium Oxide, Calcium Hydroxide, Ammonia solution, milk, Distilled water, Vinegar and Nitric acid.

From the list of substances above, pick one which is

- i) a strong base.....
- ii) an alkaline solution.....
- iii) a weak acid.....

[3]

c) **Calcium Hydroxide** and **Nitric acid** react as shown by the word equation below



i) Name the **salt** produced above.

..... [1]

ii) State the name of the process represented by the equation above.

..... [1]

iii) Briefly, describe how you can obtain dry crystals of the salt you have named above.

.....
.....
.....
.....
..... [2]

[Total: 100 Marks]

THE END